

Everest Engineering College

Sanepa, Lalitpur

Assignment 1 , 2024

Level : BE First Semester

Subject: Calculus I

Due Date: Dec 1, 2024

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. A function is defined as

$$f(x) = \begin{cases} 4x^3 - 2 & \text{if } x > 1 \\ 11 & \text{if } x = 1 \\ x^2 + 1 & \text{if } x < 1 \end{cases}.$$

Is the function $f(x)$ continuous at $x = 1$. If not why ? Restate to make it a continuous function.

2. A function

$$f(x) = \begin{cases} (1+x)^{1/x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}.$$

If the function $f(x)$ continuous at $x = 0$. If not why ? Restate to make it a continuous function.

3. If a function

$$f(x) = \begin{cases} \frac{\sin 2x}{5x} & \text{if } x \neq 0 \\ k & \text{if } x = 0 \end{cases}.$$

is continuous at $x = 0$. Find the values of k .

4. If a function

$$f(x) = \begin{cases} 3ax + b & \text{if } x > 0 \\ 11 & \text{if } x = 1 \\ 5ax - 2b & \text{if } x < 1 \end{cases}.$$

is continuous at $x = 1$. Find the values of a and b .

5. let

$$f(x) = \begin{cases} 5x - 4 & \text{if } 0 < x \leq 1 \\ 4x^2 - 3x & \text{if } 1 < x < 2 \\ 3x + 4 & \text{if } x \geq 2 \end{cases}.$$

Show that $f(x)$ continuous at $x = 1$ and $x = 2$. Also prove that $f'(x)$ exists for $x = 1$ but not at $x = 2$.

6. let

$$f(x) = \begin{cases} 2 & \text{if } x \in (-\infty, 0) \\ 2 + \sin 2x & \text{if } x \in [0, \frac{\pi}{2}) \\ 2 + (x - \frac{\pi}{2})^2 & \text{if } x \in [\frac{\pi}{2}, \infty) \end{cases}.$$

Check continuity and then differentiability at $x = 0$ and $x = \frac{\pi}{2}$.

(**Hint:** see similar problem as in example 10 on our book, page no.16)

7. Let $f(x) = |x| + |x - 2|$. Check continuity and differentiability at $x = 0$ and $x = 2$.

Hint: see similar problem as in example 11 on our book, page no. 16)

8. Evaluate: $\lim_{x \rightarrow 0} \frac{(e^x - 1)\tan^2 x}{x^3}$

9. Evaluate: $\lim_{x \rightarrow 0} \frac{xe^x - (1+x)\log(1+x)}{x^2}$

10. Evaluate: $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$

11. Evaluate: $\lim_{x \rightarrow 0} (\frac{1}{x^2} - \cot^2 x)$

12. Evaluate: $\lim_{x \rightarrow 0} (\cot^2 x)^{\sin x}$

13. Evaluate: $\lim_{x \rightarrow 0} (\sec x)^{\frac{1}{x^2}}$

14. Evaluate: $\lim_{x \rightarrow 0} (\tan 2x)^{2x}$

15. Evaluate: $\lim_{x \rightarrow 0} (\frac{\tan x}{x})^{\frac{1}{x^2}}$

16. Evaluate: $\lim_{x \rightarrow 1} x^{\frac{1}{1-x}}$

17. Evaluate: $\lim_{x \rightarrow \pi/2} (\sin x)^{\tan x}$

"ALL THE BEST."